

QUESTION_5

Aim

To analyze the time complexity of the given recurrence relation using the Master Theorem, identify the appropriate case, and evaluate the efficiency of the algorithm for large financial datasets.

Procedure

1. Consider the given recurrence relation.
2. Identify the values of a , b , and $f(n)$.
3. Compare the function $f(n)$ with the standard Master Theorem term.
4. Determine the appropriate Master Theorem case.
5. Find the asymptotic time complexity.
6. Analyze the algorithm's performance for increasing dataset sizes.
7. Conclude the efficiency of the financial analytics system.

Algorithm

Step 1: Start.

Step 2: Consider the recurrence relation:

- $T(n) = 2T(n/2) + n \log n$

Step 3: Identify the parameters:

- $a = 2$
- $b = 2$
- $f(n) = n \log n$

Step 4: Compare $f(n)$ with the standard Master Theorem term.

Step 5: Since $f(n)$ grows as $n \log n$, it satisfies Master Theorem Case 2.

Step 6: Determine the time complexity as:

- $\Theta(n(\log n)^2)$

Step 7: Analyze the algorithm for different dataset sizes and observe its growth.

Step 8: Stop.

Input

Recurrence Relation:

- $T(n) = 2T(n/2) + n \log n$

Output

- Value of $a = 2$
- Value of $b = 2$
- Function $f(n) = n \log n$
- Master Theorem Case: Case 2
- Time Complexity: $\Theta(n(\log n)^2)$

Result

The given recurrence relation satisfies Master Theorem Case 2. Therefore, the overall time complexity of the algorithm is $\Theta(n(\log n)^2)$.